<u>Michael Peña Statement of Qualification JC-528005 Consumer Research Data Analyst</u>

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I hold a Master of Science degree in Statistics from California State University, Fullerton. My education gave me strong training in quantitative research methodologies, scientific research methods, statistical modeling, and data analysis. I completed coursework in probability, statistical inference, statistical learning, Bayesian data analysis, time series analysis, and statistical computing. During my master's program, I completed several analytical and statistical projects. These projects required project design, project execution, data collection techniques, data manipulation, data aggregations, statistical significance testing, and preparing technical reports. I analyzed both structured data and unstructured data. I used analytical and statistical techniques to identify patterns, evaluate research questions, and develop meaningful information from data. One project used Cox proportional hazards regression to study housing project approval times in transit-oriented development areas in Los Angeles. I collected qualitative and quantitative data from existing research, conducted a situational analysis, and applied an appropriate statistical model to answer the research question which strengthened the evidence supporting the conclusions of a prior study. This project enhanced my ability to evaluate academic research and apply advanced statistical models to analyze complex problems through data. In another project, I analyzed more than 2.4 million quantitative electromyogram observations. The goal was to predict muscle activation patterns. The project required data manipulation, data aggregation, model development, and post-analysis review. This work involved large datasets of quantitative data. In addition, the project workflow required me to build part of the modeling procedure directly in R because available packages were not suitable for the analysis. In addition to coursework, I participated in a research project that resulted in a peer-reviewed publication. This experience required strong written and verbal communication. I worked with researchers to explain statistical methods, present data, present statistical analysis, and prepare technical reports. I regularly conveyed technical findings to a non-technical audience and explained why specific qualitative or quantitative analytical approaches were selected. Throughout my graduate program, nearly every assignment, assessment, and project required coding in R. I consistently used RStudio to conduct analyses, develop models, and prepare results. I also used Excel to review datasets, perform initial data validation, and conduct preliminary exploratory analyses before beginning more extensive work in R.

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I worked on a research project that later became the peer-reviewed publication *A Pilot Study on Aging-Related Effects on Step Performance: The Role of Muscle Quality, Size, and Strength.* The goal of the project was to determine how age-related changes in muscle quality, muscle size, and strength affect balance recovery and fall risk. The lead researcher provided raw qualitative and quantitative data from a physical therapy laboratory. The data came from Excel files and required substantial preparation before analysis. I collected, compiled, and reviewed the data. I identified missing values, inconsistent records, and observations that required verification and all of these required following up with the lead researcher and involved back and forth. I used R, RStudio, and Excel to clean, organize, and consolidate the data into a format suitable for analysis. After data preparation, I conducted exploratory data analysis. I created data tables, charts, graphs, and other visual aids to identify trends and potential relationships. I calculated descriptive summary statistics and reviewed distributions of key variables. I performed principal component analysis to reduce the number of variables and identify important patterns in the data. I evaluated several analytical approaches, including dimensionality reduction through principal component analysis, before determining that a multiple-response linear regression model was the most appropriate method for addressing the research objectives. After review and modification of the analytical plan, I developed a multiple-response linear regression model to address the research questions. Throughout the project, I held consultative meetings with the lead researcher. I presented data, presented statistical analysis, and explained complex problems in user-friendly language. I conveyed the purpose of each analytical step and explained the strengths and limitations of the selected methods. This required strong written and verbal communication and the ability to communicate with a non-technical audience. I also prepared technical reports and contributed to the manuscript. I wrote the statistical analysis section and assisted with revisions to the discussion section. The final model did not identify statistically significant relationships between the variables of interest. Based on the post-analysis findings, I recommended collecting additional observations before making definitive conclusions. The recommendation was supported by statistical evidence and the limitations of the sample size. The completed analysis became part of a peer-reviewed publication and provided a foundation for future research initiatives related to aging, balance recovery, and fall risk.